

Time-Resolved Fluorescence Spectroscopy

Introduction

Normal fluorescence is useful as a highly selective and sensitive non-invasive probe. However, better chemical information can be gained from the same experiment. While normal fluorescence spectroscopy is useful as a highly selective and sensitive non-invasive probe, better chemical information can often be gained from the same experiment by exploiting the time-dependent nature of fluorescence.

Time-resolved fluorescence provides more information about the molecular environment of the fluorophore than steady-state fluorescence measurements. Since the fluorescence lifetime of a molecule is very sensitive to its molecular environment, measurement of the fluorescence lifetime(s) reveals much about the state of the fluorophore. Many macromolecular events, such as rotational diffusion, resonance-energy transfer, and dynamic quenching, occur on the same time scale as the fluorescence decay. Thus, time-resolved fluorescence spectroscopy can be used to investigate these processes and gain insight into the chemical surroundings of the fluorophore.

It is important to remember that the fluorescence lifetime is an average time for a molecule to remain in the excited state before emitting a photon. Each individual molecule emits randomly after excitation. Many excited molecules will fluoresce before the average lifetime, but some will also fluoresce long after the average lifetime. Fluorescence lifetimes are generally on the order of 1-10 nsec, although they can range from hundreds of nanoseconds to the sub-nanosecond time scale.

Instrumentation

The instrumentation for time-resolved fluorescence spectroscopy is basically the same as for steady-state experiments. An excitation beam with a narrow wavelength range is directed at the sample, where it excites fluorescence. The fluorescence emission is collected at a 90° angle from the excitation to prevent light from the excitation source from interfering with the detection of the weaker fluorescence emission. The collected fluorescence emission enters a spectrograph and a detector registers the emission spectrum. The key differences for time-resolved spectroscopy are the replacement of the continuous light source with a pulsed source and the use of gated detection of the fluorescence emission (Figure 1).

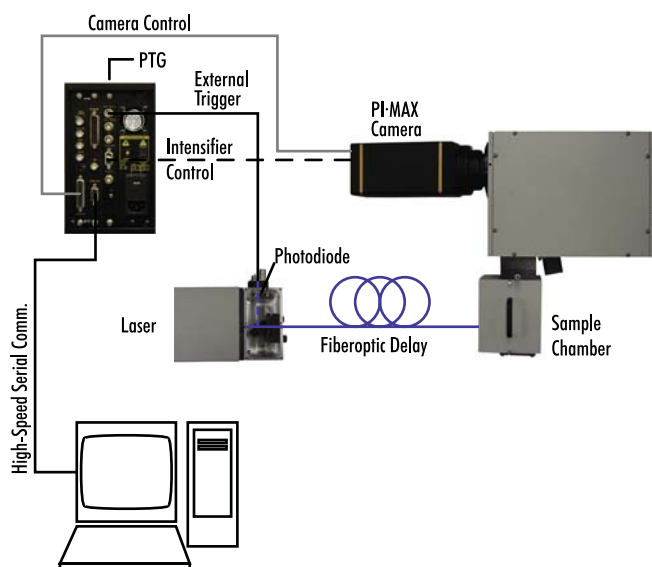


Figure 1. Experiment setup for time-resolved fluorescence spectroscopy used for this note.



TRF Spectroscopy

The two main types of pulsed-light sources used in time-resolved fluorescence spectroscopy are the flash lamp and the laser. As the variety of visible and UV pulsed lasers available on the market continues to increase, the flash lamp is disappearing from usage. The characteristics and features of pulsed lasers vary widely and the experiment setup must be tailored to a specific laser type. If the laser is free-running, then jitter will be low (jitter is the variation in the time between pulses) and a laser pre-trigger can be used to trigger the timing generator. If the time between the pre-trigger and the laser output is long enough, the timing generator can be adjusted to catch each laser pulse. If the time between the pre-trigger and the laser output is not long enough to account for all the delays between the pre-trigger and the timing generator, the delay can be adjusted to catch the subsequent laser pulse.

If the laser is a significant source of jitter, data acquisition can be triggered off the laser pulse itself rather than from the laser pre-trigger. Triggering off the laser pulse is accomplished by diverting a small percentage of the laser beam to a high-speed photodiode. The laser pulse must then be delayed long enough for the photodiode trigger to reach the timing generator, for the generator to synchronize with the incoming trigger, and for the gate pulse to reach the detector. Delay of the laser pulse can be accomplished by increasing the laser path in air or by launching the laser pulse into a fiberoptic cable of an appropriate length.

There are three key desirable features for a detector in time-resolved fluorescence spectroscopy: sensitivity, repetition rate, and response time. A high sensitivity is necessary to measure the weak signals, commonly only a few photons of fluorescence per pulse. Repetition rate is key because most pulsed light sources operate in the kHz range. If the detector cannot gate as fast or faster than the laser, pulses will be missed. In the UV, missing pulses can be a problem, since image intensifiers are sensitive to UV radiation. Even while the intensifier is off, light can reach the detector, creating unwanted signal (see MCP Bracket Pulsing™ below). Time resolution is important because with a long impulse (either laser-pulse width or intensifier-gate width), a detector with a short time resolution can distinguish the difference between the excitation pulse and the emission decay.

Analysis of Results

The goal of data analysis in time-resolved fluorescence spectroscopy is to extract the excited-state lifetime(s) from the excitation and emission data. If the fluorescence lifetime is much longer than the excitation pulse, the log of the decay curve is plotted and the lifetime is calculated from the slope of the resulting line (see Figure 2). If the fluorescence lifetime is shorter than the excitation pulse, then the decay must be deconvolved from the excitation pulse. A common algorithm for retrieving the lifetime in this case is the method of Least Squares Iterative Reconvolution. The excitation pulse is convolved with exponential decay functions of varying lifetimes until the lifetime that most closely matches the emission data is found (see Figure 3).

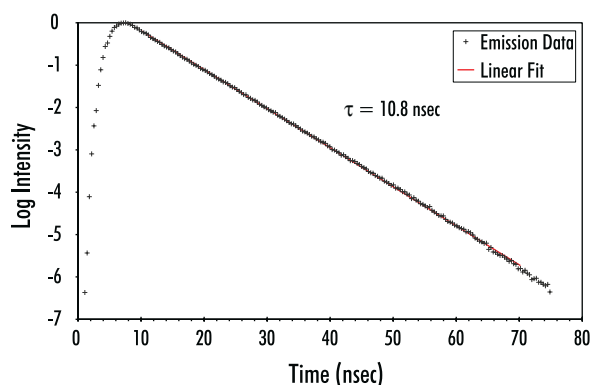


Figure 2. Log-linear plot of the time-resolved fluorescence of 9-cyanoanthracene. The slope of the line yielded a lifetime of 10.8 nsec.



TRF Spectroscopy

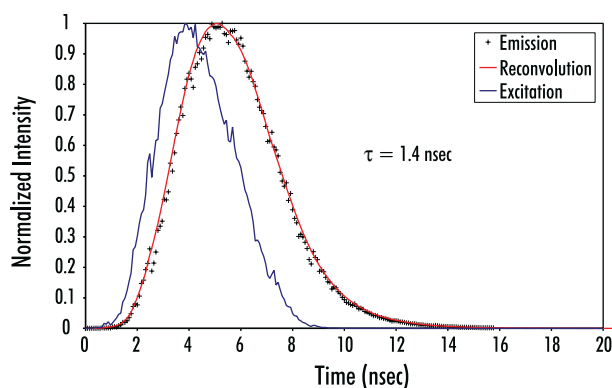


Figure 3. Plot of the excitation pulse and time-resolved fluorescence of 1,4-bis(5-phenyloxazol-2-yl)benzene (POPOP). The red curve indicates the best fit from Least Squares Iterative Reconvolution. The resulting lifetime was 1.4 nsec.

PI-MAX Cameras from Princeton Instruments

The Princeton Instruments PI-MAX™ series of intensified high-performance CCD cameras is ideally suited for time-resolved fluorescence measurements. These cameras feature fast gating to resolve short lifetimes. They have the speed to synchronize with kHz lasers and to gate on every pulse. Sensitivity is sufficient to measure photons from very weak fluorophores. Using the MCP Bracket Pulsing™ feature, UV rejection can be improved to 107:1, preventing unwanted pulses from getting through when the intensifier is gated off.

The Princeton Instruments PTG™ (Programmable Timing Generator) provides added benefits that are essential to time-resolved fluorescence spectroscopy. With 40-psec resolution, it can be used to resolve lifetimes of less than a nanosecond. The 25-nsec insertion delay allows the PI-MAX to capture pulses much faster than other pulse generators. Integration of the PTG with the PI-MAX controller allows precise on-chip accumulation to detect weak signals and to improve signal-to-noise ratios.

The WinSpec software of PI-MAX cameras is ideal for data acquisition and data reduction for time-resolved fluorescence spectroscopy. The full-software control of the spectrograph and pulse timing generator simplify the setup of complex experimental sequences. The software has an array of post-processing functions to extract excitation and emission profiles from raw spectral data. The built-in Visual Basic™ macro record greatly facilitates processing of data. WinSpec also makes it easy to export results to other data-analysis packages.



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